

TensorEngine: informe de corrida

Modelo: R_plus_alpha_times_RX

Run ID: run_c414b37057915ed9ac59

Estado: partial

Backend: structural-python 0.9.0

Verificaciones: 46 aprobadas, 0 fallidas, 2 indeterminadas.

Presentación: simplificación protegida, sin modificar los objetos canónicos. Las operaciones e hipótesis usadas por expresión constan en `presentation.json` y en los comentarios del archivo LaTeX. La validación xAct corresponde a la IR canónica.

Deltas canónicos: 1685 sustituciones y 0 trazas registradas en las pasadas del álgebra (incluidas verificaciones); 0 deltas explícitos en las 13 cantidades finales. Detalle e índices sustituidos en `delta_contractions.json`.

Expresiones tensoriales abstractas

Las expresiones de esta sección son covariantes y no incorporan sustituciones del ansatz.

L

$$L = R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3} u_{d_4} g^{d_4 d_5} u_{d_5} \alpha + R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3}$$

Significado: Densidad lagrangiana escalar normalizada de la teoría.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: validated_model

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

M_{ab}

$$M_{ab} = 2 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} \alpha + 2 R_{a d_0 b}{}^{d_0} + u_a u_b R_{d_0 d_1}{}^{d_0 d_1} \alpha$$

Significado: Derivada de L respecto de la métrica inversa, M_{ab} .

Índices libres y varianzas: $a_{\downarrow}, b_{\downarrow}$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

P^{abcd}

$$P^{abcd} = -\frac{g^{a d} g^{b c} u_{d_0} u^{d_0} \alpha}{2} - \frac{g^{a d} g^{b c}}{2} + \frac{g^{a c} g^{b d} u_{d_0} u^{d_0} \alpha}{2} + \frac{g^{a c} g^{b d}}{2}$$

Significado: Momento de curvatura $P^{abcd} = \partial L / \partial R_{abcd}$.

Índices libres y varianzas: $a^{\uparrow}, b^{\uparrow}, c^{\uparrow}, d^{\uparrow}$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

J^a

$$J^a = 2 u^a R_{d_0 d_1}{}^{d_0 d_1} \alpha$$

Significado: Momento J^a conjugado a $\nabla_a \phi$.

Índices libres y varianzas: $a^\uparrow$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

F_ϕ

$$F_\phi = 0$$

Significado: Derivada explícita $F_\phi = \partial L / \partial \phi$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

E_{ab}

$$\begin{aligned} E_{ab} = & -\nabla_a \left(u^{d_0} \right) \nabla_{d_0} (u_b) \alpha - \nabla_b \left(u^{d_0} \right) \nabla_{d_0} (u_a) \alpha + 2 \nabla_{d_1} (u_{d_0}) \nabla_{d_2} \left(u^{d_1} \right) g_{ab} g^{d_0 d_2} \alpha \\ & - \frac{\nabla_a (\nabla_b (u^{d_0})) u_{d_0} \alpha}{2} - \frac{\nabla_a (\nabla_{d_0} (u_b)) u^{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_a (u^{d_0})) u_{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_{d_0} (u_a)) u^{d_0} \alpha}{2} \\ & - \frac{g_{ab} R_{d_0 d_1}{}^{d_0 d_1} u_{d_2} u^{d_2} \alpha}{2} - \frac{g_{ab} R_{d_0 d_1}{}^{d_0 d_1}}{2} + \nabla_{d_2} (\nabla_{d_1} (u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} \alpha \\ & + \nabla_{d_2} \left(\nabla_{d_1} \left(u^{d_0} \right) \right) g_{ab} u_{d_0} g^{d_1 d_2} \alpha + R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} \alpha + u_a u_b R_{d_0 d_1}{}^{d_0 d_1} \alpha + R_{a d_0 b}{}^{d_0} \end{aligned}$$

Significado: Ecuación de Euler-Lagrange métrica E_{ab} .

Índices libres y varianzas: $a_\downarrow, b_\downarrow$

Fuentes del pipeline: delta_lagrangian, curvature_derivative_metric_term

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$$\begin{aligned} E_{ab}|_{\nabla \nabla P} = & -\nabla_a \left(u^{d_0} \right) \nabla_{d_0} (u_b) \alpha - \nabla_b \left(u^{d_0} \right) \nabla_{d_0} (u_a) \alpha + 2 \nabla_{d_1} (u_{d_0}) \nabla_{d_2} \left(u^{d_1} \right) g_{ab} g^{d_0 d_2} \alpha \\ & - \frac{\nabla_a (\nabla_b (u^{d_0})) u_{d_0} \alpha}{2} - \frac{\nabla_a (\nabla_{d_0} (u_b)) u^{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_a (u^{d_0})) u_{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_{d_0} (u_a)) u^{d_0} \alpha}{2} \\ & + \nabla_{d_2} (\nabla_{d_1} (u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} \alpha + \nabla_{d_2} \left(\nabla_{d_1} \left(u^{d_0} \right) \right) g_{ab} u_{d_0} g^{d_1 d_2} \alpha \end{aligned}$$

La expresión anterior identifica la contribución $-2\nabla^c \nabla^d P_{acdb}$ dentro de E_{ab} .

E_ϕ

$$E_\phi = -2\alpha \left(\nabla_{d_0} \left(u^{d_0} \right) R_{d_1 d_2}{}^{d_1 d_2} + \nabla_{d_2} \left(R_{d_0 d_1}{}^{d_0 d_1} \right) u^{d_2} \right)$$

Significado: Ecuación de Euler-Lagrange escalar E_ϕ .

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: scalar_derivative, scalar_gradient_momentum

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\mathcal{R}$

$$\mathcal{R} = g^{ac} g^{bd} R_{abcd}$$

Significado: Escalar de Ricci construido con la convención geométrica activa.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: validated_model

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

$$R_{ab}R^{ab} = g^{ac} R_{abcd} g^{be} g^{df} g^{gh} R_{g e h f}$$

Significado: Invariante cuadrático de Ricci $R_{ab} R^{ab}$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: validated_model, riemann_tensor

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

$$R_{abcd} = R_{a b c d}$$

Significado: Tensor de Riemann covariante tratado como entrada geométrica.

Índices libres y varianzas: $a_\downarrow, b_\downarrow, c_\downarrow, d_\downarrow$

Fuentes del pipeline: validated_model

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{abcd}R^{abcd}$

$$R_{abcd}R^{abcd} = g^{ae} g^{bf} g^{cg} g^{dh} R_{a b c d} R_{e f g h}$$

Significado: Invariante de Kretschmann $R_{abcd} R^{abcd}$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: validated_model, riemann_tensor

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_e P^{abcd}$$

$$\nabla_e P^{abcd} = \frac{1}{2} \alpha \left(\nabla_{d_0}(u_e) g^{ac} g^{bd} u^{d_0} + \nabla_e(u^{d_0}) g^{ac} g^{bd} u_{d_0} - \nabla_{d_0}(u_e) g^{ad} g^{bc} u^{d_0} - \nabla_e(u^{d_0}) g^{ad} g^{bc} u_{d_0} \right)$$

Significado: Primera derivada covariante del momento de curvatura.

Índices libres y varianzas: $a^\uparrow, b^\uparrow, c^\uparrow, d^\uparrow, e_\downarrow$

Fuentes del pipeline: curvature_momentum

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_f \nabla_e P^{abcd}$$

$$\begin{aligned} \nabla_f \nabla_e P^{abcd} = \frac{1}{2} \alpha \left(\nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ac} g^{bd} + \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ac} g^{bd} + \nabla_f(\nabla_{d_0}(u_e)) g^{ac} g^{bd} u^{d_0} \right. \\ \left. + \nabla_f(\nabla_e(u^{d_0})) g^{ac} g^{bd} u_{d_0} - \nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ad} g^{bc} - \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ad} g^{bc} \right. \\ \left. - \nabla_f(\nabla_{d_0}(u_e)) g^{ad} g^{bc} u^{d_0} - \nabla_f(\nabla_e(u^{d_0})) g^{ad} g^{bc} u_{d_0} \right) \end{aligned}$$

Significado: Segunda derivada covariante del momento de curvatura.

Índices libres y varianzas: $a^\uparrow, b^\uparrow, c^\uparrow, d^\uparrow, e_\downarrow, f_\downarrow$

Fuentes del pipeline: nabla_P

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

Descomposición compacta adicional

Esta vista se agrega después de los resultados anteriores y reutiliza exactamente la IR canónica ya calculada. No interviene en la validación ni en los fingerprints.

Descomposición de E_{ab} .

$$E_{ab} = M_{ab} - \mathcal{R}_{(ab)}[P] - \frac{1}{2} g_{ab} L - 2 \nabla_m \nabla_n P^{mn}_{ab}$$

Reconstrucción IR: verified. Los cuatro bloques reconstruyen exactamente E_{ab} en la IR canónica.

forma compacta:
$$\begin{aligned} E_{ab} = & -\nabla_a(u^{d_0}) \nabla_{d_0}(u_b) \alpha - \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) \alpha \\ & + 2 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{ab} g^{d_0 d_2} \alpha - \frac{\nabla_a(\nabla_b(u^{d_0})) u_{d_0} \alpha}{2} \\ & - \frac{\nabla_a(\nabla_{d_0}(u_b)) u^{d_0} \alpha}{2} - \frac{\nabla_b(\nabla_a(u^{d_0})) u_{d_0} \alpha}{2} \\ & - \frac{\nabla_b(\nabla_{d_0}(u_a)) u^{d_0} \alpha}{2} - \frac{g_{ab} R_{d_0 d_1}^{d_0 d_1} u_{d_2} u^{d_2} \alpha}{2} - \frac{g_{ab} R_{d_0 d_1}^{d_0 d_1}}{2} \\ & + \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} \alpha + \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{ab} u_{d_0} g^{d_1 d_2} \alpha \\ & + R_{a d_0 b}^{d_0} u_{d_1} u^{d_1} \alpha + u_a u_b R_{d_0 d_1}^{d_0 d_1} \alpha + R_{a d_0 b}^{d_0} \end{aligned}$$

Bloque M_{ab} (fuentes: metric_momentum).

forma compacta:
$$M_{ab} = 2 R_{a d_0 b}^{d_0} u_{d_1} u^{d_1} \alpha + 2 R_{a d_0 b}^{d_0} + u_a u_b R_{d_0 d_1}^{d_0 d_1} \alpha$$

Bloque $-\mathcal{R}_{(ab)}[P]$ (fuentes: curvature_momentum, riemann_tensor).

forma compacta:
$$-\mathcal{R}_{(ab)}[P] = \frac{1}{2} \left(-g_{a d_1} \left(-\frac{g^{d_1 d_2} g^{d_3 d_4} u_{d_0} u^{d_0} \alpha}{2} - \frac{g^{d_1 d_2} g^{d_3 d_4}}{2} \right. \right. \\ \left. \left. + \frac{g^{d_1 d_4} g^{d_3 d_2} u_{d_0} u^{d_0} \alpha}{2} + \frac{g^{d_1 d_4} g^{d_3 d_2}}{2} \right) R_{b d_3 d_4 d_2} \right. \\ \left. - g_{b d_1} \left(-\frac{g^{d_1 d_2} g^{d_3 d_4} u_{d_0} u^{d_0} \alpha}{2} - \frac{g^{d_1 d_2} g^{d_3 d_4}}{2} + \frac{g^{d_1 d_4} g^{d_3 d_2} u_{d_0} u^{d_0} \alpha}{2} \right. \right. \\ \left. \left. + \frac{g^{d_1 d_4} g^{d_3 d_2}}{2} \right) R_{a d_3 d_4 d_2} \right)$$

Bloque $-\frac{1}{2}g_{ab}L$ (fuentes: `lagrangian`).

forma compacta:
$$-\frac{1}{2}g_{ab}L = -\frac{1}{2}g_{a b} \left(R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3} u_{d_4} g^{d_4 d_5} u_{d_5} \alpha + R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3} \right)$$

Bloque $E_{ab}^{(\nabla\nabla P)}$ (fuentes: `curvature_momentum`, `nabla_nabla_P`).

forma compacta:
$$E_{ab}^{(\nabla\nabla P)} = -\nabla_a \left(u^{d_0} \right) \nabla_{d_0} (u_b) \alpha - \nabla_b \left(u^{d_0} \right) \nabla_{d_0} (u_a) \alpha \\ + 2 \nabla_{d_1} (u_{d_0}) \nabla_{d_2} \left(u^{d_1} \right) g_{a b} g^{d_0 d_2} \alpha - \frac{\nabla_a \left(\nabla_b (u^{d_0}) \right) u_{d_0} \alpha}{2} \\ - \frac{\nabla_a (\nabla_{d_0} (u_b)) u^{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_a (u^{d_0})) u_{d_0} \alpha}{2} - \frac{\nabla_b (\nabla_{d_0} (u_a)) u^{d_0} \alpha}{2} \\ + \nabla_{d_2} (\nabla_{d_1} (u_{d_0})) g_{a b} g^{d_0 d_2} u^{d_1} \alpha + \nabla_{d_2} \left(\nabla_{d_1} \left(u^{d_0} \right) \right) g_{a b} u_{d_0} g^{d_1 d_2} \alpha$$

Descomposición de E_ϕ .

$$E_\phi = F_\phi - \nabla_a J^a$$

Reconstrucción IR: verified. Los dos bloques reconstruyen exactamente E_ϕ en la IR canónica.

forma compacta:
$$E_\phi = -2 \alpha \left(\nabla_{d_0} \left(u^{d_0} \right) R_{d_1 d_2}{}^{d_1 d_2} + \nabla_{d_2} \left(R_{d_0 d_1}{}^{d_0 d_1} \right) u^{d_2} \right)$$

Bloque F_ϕ (fuentes: `scalar_derivative`).

forma compacta:
$$F_\phi = 0$$

Bloque $-\nabla_a J^a$ (fuentes: `scalar_gradient_momentum`, `scalar_euler`).

forma compacta:
$$-\nabla_a J^a = -2 \alpha \left(\nabla_{d_0} \left(u^{d_0} \right) R_{d_1 d_2}{}^{d_1 d_2} + \nabla_{d_2} \left(R_{d_0 d_1}{}^{d_0 d_1} \right) u^{d_2} \right)$$

Descomposición de P^{abcd} .

$$P^{abcd} = \partial L / \partial R_{abcd}$$

Reconstrucción IR: verified. La forma compacta y la expandida referencian el mismo objeto P^{abcd} .

$$\text{forma compacta: } P^{abcd} = -\frac{g^{ad} g^{bc} u_{d0} u^{d0} \alpha}{2} - \frac{g^{ad} g^{bc}}{2} + \frac{g^{ac} g^{bd} u_{d0} u^{d0} \alpha}{2} + \frac{g^{ac} g^{bd}}{2}$$

Expresiones proyectadas mediante el ansatz

Ansatz utilizado: draft4_circular.

$$\begin{aligned} (x^\mu) &= (\tau, r, \varphi), \quad ds^2 \\ &= -f(r) d\tau^2 + \frac{1}{f(r)} dr^2 + r^2 d\varphi^2 \end{aligned}$$

$$\phi = \Phi(r, \varphi) \quad (\text{genérico, sin especialización})$$

Las componentes completas se conservan de forma dispersa en **results.json**; el documento muestra como máximo doce componentes no nulas por tensor.

L

Ansatz: draft4_circular; **estado:** completada.

$$\begin{aligned} L|_{\text{ansatz}} = -\frac{1}{r^3} & \left(\alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 (2 f'(r) + r f''(r)) + f''(r) r^3 \left(\alpha f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 + 1 \right) \right. \\ & \left. + 2 f'(r) r^2 \left(\alpha f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 + 1 \right) \right) \end{aligned}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'.

M_{ab}

Ansatz: draft4_circular; **estado:** completada.

$$[M_{ab}]_{00} = \frac{1}{r^3} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) (r f''(r) + f'(r)) f(r)$$

$$\begin{aligned} [M_{ab}]_{11} = -\frac{1}{r^3 f(r)} & \left(r^2 (r f''(r) + f'(r)) + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) (r f''(r) + f'(r)) \right. \\ & \left. + \alpha r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 (2 f'(r) + r f''(r)) f(r) \right) \end{aligned}$$

$$[M_{ab}]_{12} = -\frac{1}{r} \alpha (2 f'(r) + r f''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[M_{ab}]_{21} = -\frac{1}{r} \alpha \left(2 f'(r) + r f''(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[M_{ab}]_{22} = -2 r f'(r) - \alpha f''(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 - \frac{4 f'(r) \alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{r} - 2 r f'(r) \alpha f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2$$

Proyección completa almacenada: 5 componentes no nulas de 9. Proyectada respetando el ansatz 'draft4_circular'.

P_{abcd}

Ansatz: draft4_circular; **estado:** completada.

$$\left[P_{abcd} \right]^{0110} = \frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{0220} = \frac{1}{2 r^4 f(r)} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{0101} = -\frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{0202} = -\frac{1}{2 r^4 f(r)} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{1010} = -\frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{1001} = \frac{1}{2 r^2} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P_{abcd} \right]^{1221} = -\frac{1}{2 r^4} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) f(r)$$

$$\left[P_{abcd} \right]^{1212} = \frac{1}{2 r^4} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) f(r)$$

$$\left[P_{abcd} \right]^{2020} = -\frac{1}{2 r^4 f(r)} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P^{abcd}\right]^{2121} = \frac{1}{2r^4} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) f(r)$$

$$\left[P^{abcd}\right]^{2002} = \frac{1}{2r^4 f(r)} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P^{abcd}\right]^{2112} = -\frac{1}{2r^4} \left(r^2 + \alpha \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right) f(r)$$

Proyección completa almacenada: 12 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'.

J^a

Ansatz: draft4_circular; **estado:** completada.

$$[J^a]^1 = -\frac{2\alpha (2f'(r) + rf''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f(r)}{r}$$

$$[J^a]^2 = -\frac{2\alpha (2f'(r) + rf''(r)) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)}{r^3}$$

Proyección completa almacenada: 2 componentes no nulas de 3. Proyectada respetando el ansatz 'draft4_circular'.

F_ϕ

Ansatz: draft4_circular; **estado:** completada.

$$F_\phi|_{\text{ansatz}} = 0$$

Proyección completa almacenada: 0 componentes no nulas de 1. La expresión abstracta es nula; todas sus componentes son cero.

E_{ab}

Ansatz: draft4_circular; **estado:** completada.

$$\begin{aligned} [E_{ab}]_{00} = \frac{1}{2r^4} f(r) & \left(\alpha \left(-4 \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi)^2 - 8f(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 - 4 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial \varphi^3}(r, \varphi) \right. \right. \\ & + r f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 - r^4 f'(r)^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 - 8f(r) r^2 \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi)^2 - 4r^4 f(r)^2 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi)^2 \\ & - 4 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f(r) \frac{\partial^3 \Phi}{\partial r^2 \partial \varphi^1}(r, \varphi) r^2 - 4f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^1 \partial \varphi^2}(r, \varphi) r^2 \\ & - 4 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) r^3 f(r)^2 - 4 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^3}(r, \varphi) r^4 f(r)^2 \\ & - 3f'(r) f(r) r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 - 2f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) r^2 - 2f(r) f''(r) r^4 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \\ & \left. \left. + 12r \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f(r) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) - 10f'(r) f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) r^4 \right) - f'(r) r^3 \right) \end{aligned}$$

$$\begin{aligned}
[E_{ab}]_{11} = & \frac{1}{2r^4 f(r)} \left(f'(r) r^3 \right. \\
& + \alpha \left(4 \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi)^2 - 4 f(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 4 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial \varphi^3}(r, \varphi) + r^4 f'(r)^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \right. \\
& \left. - r f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 4 f(r) r^2 \left(\frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi)^2 + \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^1 \partial \varphi^2}(r, \varphi) \right) \right. \\
& + f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) r^3 \left(-f'(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) + 4 f(r) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) \right) \\
& + 2 f(r) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) r^4 \left(f'(r) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) - \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f''(r) \right) \\
& \left. + 2 f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) r^2 + 4 r f(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \Bigg)
\end{aligned}$$

$$\begin{aligned}
[E_{ab}]_{12} = & \frac{1}{r^3} \alpha \left(6 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) - 2 r \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^1 \partial \varphi^2}(r, \varphi) - 2 r \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right. \\
& - r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f''(r) - 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) \\
& - 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) - 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^2 \partial \varphi^1}(r, \varphi) f(r) \\
& \left. - 2 r^3 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f(r) + 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f(r) \right)
\end{aligned}$$

$$\begin{aligned}
[E_{ab}]_{21} = & \frac{1}{r^3} \alpha \left(6 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) - 2 r \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^1 \partial \varphi^2}(r, \varphi) - 2 r \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right. \\
& - r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f''(r) - 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) \\
& - 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) - 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^3 \Phi}{\partial r^2 \partial \varphi^1}(r, \varphi) f(r) \\
& \left. - 2 r^3 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f(r) + 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f(r) \right)
\end{aligned}$$

$$\begin{aligned}
[E_{ab}]_{22} = & \frac{1}{2r^2} \left(r^2 \left(f''(r) r^2 \right. \right. \\
& + \alpha \left(-f''(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 4 f(r) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi)^2 + 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r)^2 + 4 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f'(r) + 4 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right. \\
& \left. + 4 \alpha \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \left(-2 r \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) + 2 \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) f(r) \right) + 3 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f(r) \right) \right) \Bigg)
\end{aligned}$$

Proyección completa almacenada: 5 componentes no nulas de 9. Reutilizada desde la etapa de componentes de E_{ab}.

E_ϕ

Ansatz: draft4_circular; **estado:** completada.

$$E_\phi|_{\text{ansatz}} = \frac{1}{r^3} 2\alpha \left((2f'(r) + r f''(r)) \left(r^2 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) + \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) + r \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f(r) \right. \right. \\ \left. \left. + \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) \right) + r \left(-2f'(r) + r^2 f^{(3)}(r) + 2r f''(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f(r) \right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Reutilizada desde la etapa de componentes de E_phi.

$\mathcal{R}$

Ansatz: draft4_circular; **estado:** completada.

$$\mathcal{R}|_{\text{ansatz}} = - \left(\frac{2f'(r)}{r} + f''(r) \right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

Ansatz: draft4_circular; **estado:** completada.

$$R_{ab}R^{ab}|_{\text{ansatz}} = \frac{f''(r)^2}{2} + \frac{3f'(r)^2}{2r^2} + \frac{f'(r) f''(r)}{r}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

Ansatz: draft4_circular; **estado:** completada.

$$[R_{abcd}]_{0101} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{0110} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{0202} = \frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{0220} = -\frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{1001} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{1010} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{1212} = -\frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{1221} = \frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{2002} = -\frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{2020} = \frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{2112} = \frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{2121} = -\frac{r f'(r)}{2 f(r)}$$

Proyección completa almacenada: 12 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$R_{abcd}R^{abcd}$$

Ansatz: draft4_circular; **estado:** completada.

$$R_{abcd}R^{abcd}|_{\text{ansatz}} = f''(r)^2 + \frac{2 f'(r)^2}{r^2}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_e P^{abcd}$$

Ansatz: draft4_circular; **estado:** completada.

$$\begin{aligned} \left[\nabla_e P^{abcd} \right]_1^{0110} &= \frac{1}{2 r^3} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right. \\ &\quad \left. + r^3 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) + 2 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right) \end{aligned}$$

$$\begin{aligned} \left[\nabla_e P^{abcd} \right]_1^{0220} &= \frac{1}{2 r^5 f(r)} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right. \\ &\quad \left. + r^3 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) + 2 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right) \end{aligned}$$

$$\begin{aligned} \left[\nabla_e P^{abcd} \right]_1^{2112} &= -\frac{1}{2r^5} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right. \\ &\quad \left. + r^3 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) + 2 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right) f(r) \end{aligned}$$

Representación compacta: se muestran 12 de 24 componentes no nulas. Proyección completa almacenada: 24 componentes no nulas de 243. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_f \nabla_e P^{abcd}$$

Ansatz: draft4_circular; **estado:** completada.

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{0110} &= -\frac{1}{4r^3} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) f(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{0220} &= -\frac{1}{4r^5} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{0101} &= \frac{1}{4r^3} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) f(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{0202} &= \frac{1}{4r^5} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{1010} &= \frac{1}{4r^3} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) f(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{1001} &= -\frac{1}{4r^3} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) f(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{1221} &= \frac{1}{4r^5} \alpha f(r)^2 \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{1212} &= -\frac{1}{4r^5} \alpha f(r)^2 \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{2020} &= \frac{1}{4r^5} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{2121} &= -\frac{1}{4r^5} \alpha f(r)^2 \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{2002} &= -\frac{1}{4r^5} \alpha \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

$$\begin{aligned} \left[\nabla_f \nabla_e P^{abcd} \right]_{00}^{2112} &= \frac{1}{4r^5} \alpha f(r)^2 \left(2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r \frac{\partial^2 \Phi}{\partial r^1 \partial \varphi^1}(r, \varphi) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) \right. \\ &\quad \left. + 2 r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f(r) \right) f'(r) \end{aligned}$$

Representación compacta: se muestran 12 de 60 componentes no nulas. Proyección completa almacenada: 60 componentes no nulas de 729. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

Extras: perfiles escalares de una variable, con la misma métrica del ansatz genérico. Se sustituyen únicamente las componentes de L y P^{abcd} ya calculadas; sin validación xAct independiente de estos perfiles. Las componentes completas de estos extras constan en **presentation.json**.

Extra: $\phi = \Phi(r)$

$$L = -\frac{\left(\alpha \Phi'(r)^2 f(r) + 1\right) (2 f'(r) + r f''(r))}{r}$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd}\right]^{0101} = -\frac{\alpha f(r) \Phi'(r)^2}{2} - \frac{1}{2}$$

$$\left[P^{abcd}\right]^{0202} = -\frac{\left(\alpha \Phi'(r)^2 f(r) + 1\right)}{2 r^2 f(r)}$$

$$\left[P^{abcd}\right]^{1212} = \frac{\left(\alpha \Phi'(r)^2 f(r) + 1\right) f(r)}{2 r^2}$$

Extra: $\phi = \Phi(\varphi)$

$$L = -\frac{1}{r^3} \left(f''(r) r^3 + 2 f'(r) r^2 + \alpha \Phi'(\varphi)^2 (2 f'(r) + r f''(r)) \right)$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd}\right]^{0101} = -\frac{\left(r^2 + \alpha \Phi'(\varphi)^2\right)}{2 r^2}$$

$$\left[P^{abcd}\right]^{0202} = -\frac{\left(r^2 + \alpha \Phi'(\varphi)^2\right)}{2 r^4 f(r)}$$

$$\left[P^{abcd}\right]^{1212} = \frac{\left(r^2 + \alpha \Phi'(\varphi)^2\right) f(r)}{2 r^4}$$

Política de lectura. Este PDF/LaTeX es una vista. El archivo `results.json` conserva la representación intermedia canónica y la trazabilidad completa.